Module 13

Section 8.4

Vectors

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Vectors

5 ALEKS Skills (4 from Section 8.4 + 1 Supplementary)

Module 13 — Skills Overview

1. Writing a vector in component form from initial/terminal points
2. Vector addition and scalar multiplication: component form
3. Finding magnitude and direction of a vector from its graph
4. Finding the direction angle of a vector in ai + bj form
5. Finding the components of a vector from its graph

Vectors and Scalars

Scalar

Magnitude only (a single number)

Examples: speed, temperature, mass

Vector

Magnitude AND direction

Examples: velocity, force, displacement

Notation

Component form: $\mathbf{v} = \langle a, b \rangle$ or $\mathbf{v} = a\mathbf{i} + b\mathbf{j}$

Magnitude: $|\mathbf{v}| = \sqrt{a^2 + b^2}$

Direction angle: θ **measured counterclockwise from positive x-axis**

Two vectors are equal $\Leftrightarrow$ same components

Writing a Vector in Component Form

Formula: $v = \langle x_2 - x_1, y_2 - y_1 \rangle$ (terminal minus initial)

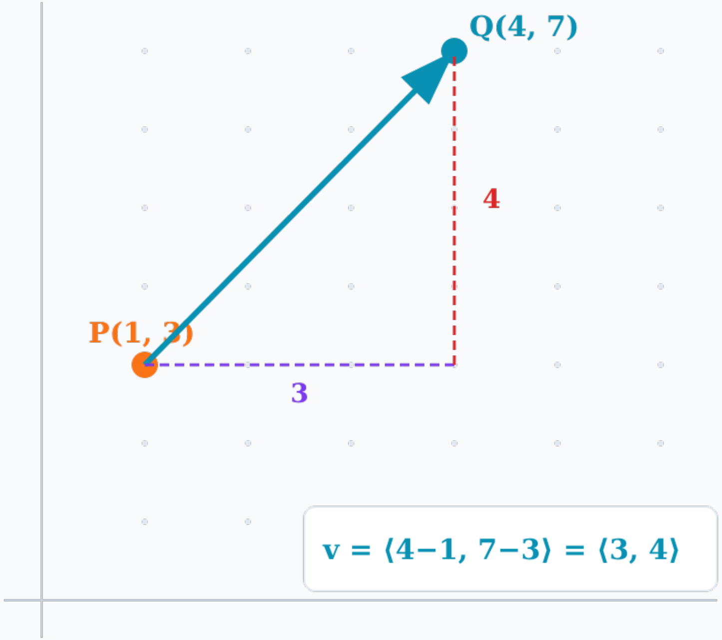
Example: A(−2, 5) to B(3, −1)

$v = \langle 3 - (-2), -1 - 5 \rangle$

$v = \langle 5, -6 \rangle$

Meaning: 5 units right, 6 units down

Component form is position-independent — only the displacement matters.



Finding Components from a Graph

Method 1: From Points

- 1. Read tail (initial) coordinates
- 2. Read tip (terminal) coordinates
- 3. Subtract: $\langle x\_tip - x\_tail, y\_tip - y\_tail \rangle$

Method 2: From |v| and θ

Horizontal: $v_x = |v| \cos\theta$

Vertical: $v_y = |v| \sin\theta$

$$\mathbf{v} = \langle |\mathbf{v}| \cos\theta, |\mathbf{v}| \sin\theta \rangle$$

Example: Vector from (1, 2) to (−3, 5)

$v = \langle -3 - 1, 5 - 2 \rangle = \langle -4, 3 \rangle$

$|v| = \sqrt{(16 + 9)} = 5$

$\theta = \tan^{-1}(3/-4) \approx -36.9° \rightarrow$ Q II: $+180° \rightarrow \theta \approx 143.1°$

Vector Addition & Scalar Multiplication

Addition	$\langle a, b\rangle + \langle c, d\rangle = \langle a+c, b+d\rangle$
Subtraction	$\langle a, b\rangle - \langle c, d\rangle = \langle a-c, b-d\rangle$
Scalar ×	$k\langle a, b\rangle = \langle ka, kb\rangle$

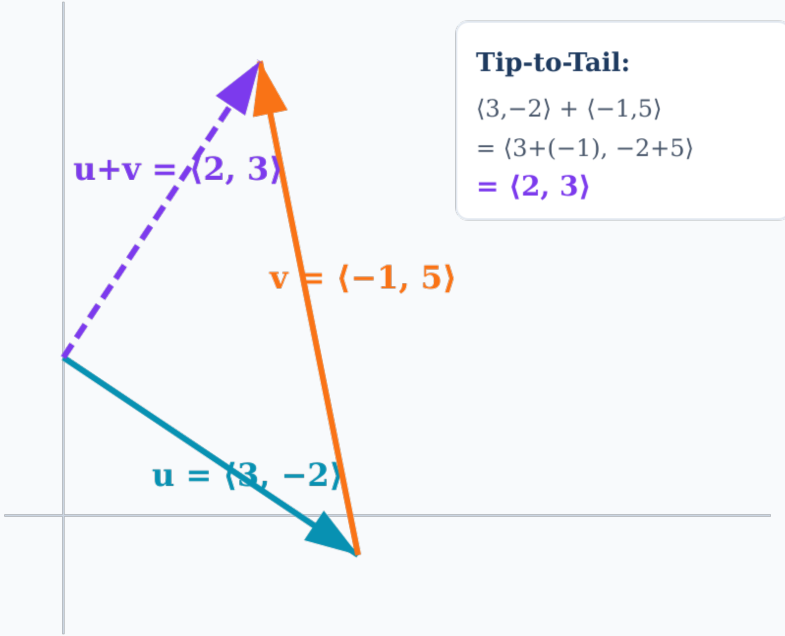
$\mathbf{u} = \langle 3, -2\rangle$ and $\mathbf{v} = \langle -1, 5\rangle$

$u + v = \langle 3+(-1), -2+5\rangle = \langle 2, 3\rangle$

3u – 2v:

$3\langle 3,-2\rangle - 2\langle -1,5\rangle = \langle 9,-6\rangle - \langle -2,10\rangle$

$= \langle 9-(-2), -6-10\rangle = \langle 11, -16\rangle$



Tip-to-Tail:

$\langle 3,-2\rangle + \langle -1,5\rangle$

$= \langle 3+(-1), -2+5\rangle$

$= \langle 2, 3\rangle$

Magnitude and Direction

Magnitude: $|v| = \sqrt{a^2 + b^2}$

Direction: $\theta = \tan^{-1}(b / a)$ (adjust for quadrant)

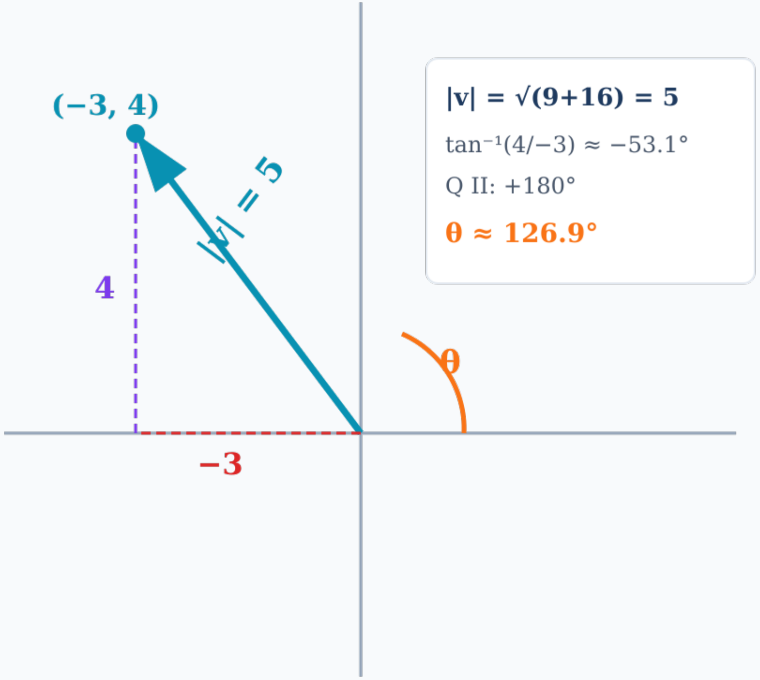
Example: $v = \langle -3, 4 \rangle$

$|v| = \sqrt{9 + 16} = \sqrt{25} = 5$

$\tan^{-1}(4/-3) \approx -53.1°$

Q II ($a < 0$, $b > 0$): add 180°

$\theta \approx -53.1° + 180° = 126.9°$



Direction Angle — Quadrant Adjustment

Q II

$\theta = \tan^{-1}(b/a) + 180°$

$a < 0, b > 0$

Q I

$\theta = \tan^{-1}(b/a)$

$a > 0, b > 0$

Q III

$\theta = \tan^{-1}(b/a) + 180°$

$a < 0, b < 0$

Q IV

$\theta = \tan^{-1}(b/a) + 360°$

$a > 0, b < 0$

Examples

$\langle 3, 4 \rangle$ Q I

$\theta \approx 53.1°$

$\langle -5, 12 \rangle$ Q II

$\theta \approx 112.6°$

$\langle -3, -7 \rangle$ Q III

$\theta \approx 246.8°$

$\langle 4, -2 \rangle$ Q IV

$\theta \approx 333.4°$

Direction Angle of ai + bj

Find the direction angle of $v = -5i + 12j$.

Convert to components:

$v = -5i + 12j \to \langle -5, 12 \rangle \quad (a = -5, b = 12)$

Find magnitude:

$|v| = \sqrt{(25 + 144)} = \sqrt{169} = 13$

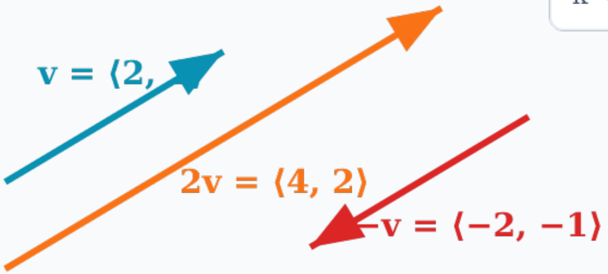
Find direction:

$\tan^{-1}(12/-5) \approx -67.4°$

$a < 0, b > 0 \to$ Q II: add 180°

$$\theta \approx -67.4° + 180° = 112.6°$$

Scalar Multiplication — Visual



Scalar k:

$k > 1 \rightarrow$ stretches

$0 < k < 1 \rightarrow$ shrinks

$k < 0 \rightarrow$ reverses

Rules

$|k| > 1 \rightarrow$ stretches the vector

$0 < |k| < 1 \rightarrow$ shrinks the vector

$k < 0 \rightarrow$ reverses direction

$$k\langle a, b\rangle = \langle ka, kb\rangle$$

Magnitude: $|kv| = |k| \times |v|$

Test Prep — Vector Operations

Test 3, Problem 9: Vector addition, scalar multiplication, magnitude, and direction angle calculations.

Part A: Component form

Given: $P(3, -1)$ to $Q(-2, 4)$

$v = \langle -2-3, 4-(-1) \rangle$

$$\mathbf{v} = \langle -5, 5 \rangle$$

$|v| = \sqrt{(25+25)} = \sqrt{50} = 5\sqrt{2}$

$\theta = \tan^{-1}(5/-5)+180° = -45°+180°$

$$\theta = 135°$$

Part B: Operations

Given: $u = \langle 2, -3 \rangle$, $v = \langle -1, 4 \rangle$

$$\mathbf{u + v = \langle 1, 1 \rangle}$$

$2u - 3v =$

$$\langle 4,-6 \rangle - \langle -3,12 \rangle = \langle 7, -18 \rangle$$

$|2u - 3v| = \sqrt{(49+324)} = \sqrt{373}$

$$\approx 19.3$$

Module 13 Summary

5 ALEKS Topics | Section 8.4

Components

$v = \langle x_2 - x_1, y_2 - y_1 \rangle$ from points, or $v = \langle |v|\cos\theta, |v|\sin\theta \rangle$ from magnitude/direction

Operations

Add/subtract component-wise, scalar multiplication multiplies each component

Magnitude & Direction

$|v| = \sqrt{a^2+b^2}$, $\theta = \tan^{-1}(b/a)$ adjusted for quadrant (Q II/III: +180°, Q IV: +360°)

Key for Test 3

Problem 9: component form, vector operations ($2u - 3v$), magnitude, and direction angle with quadrant adjustment

Next: Unit 3 Review → Test 3